How to Trade Factor Momentum in Practice

A Market-Neutral Approach for Dynamic Factor Investing

CONCRETUM RESEARCH
JUL 24, 2026 · PAID

Last week, drawing on a very insightful piece by **Man AHL** (*"A Trend Following Deep Dive: Cash (Equities) Is King"*), we investigated the presence of both **cross-sectional and time-series momentum effects** within a basket of fifteen US factor portfolios, constructed using Professor Kenneth French's publicly available data library.

Timing Equity Factors with Momentum
CONCRETUM RESEARCH · JUL 18
Read full story

While our findings are consistent with those of Panjabi, Bordigoni, and Buchanan (2026), they should not be viewed as direct confirmation that such actively managed factors can be traded directly as constructed.

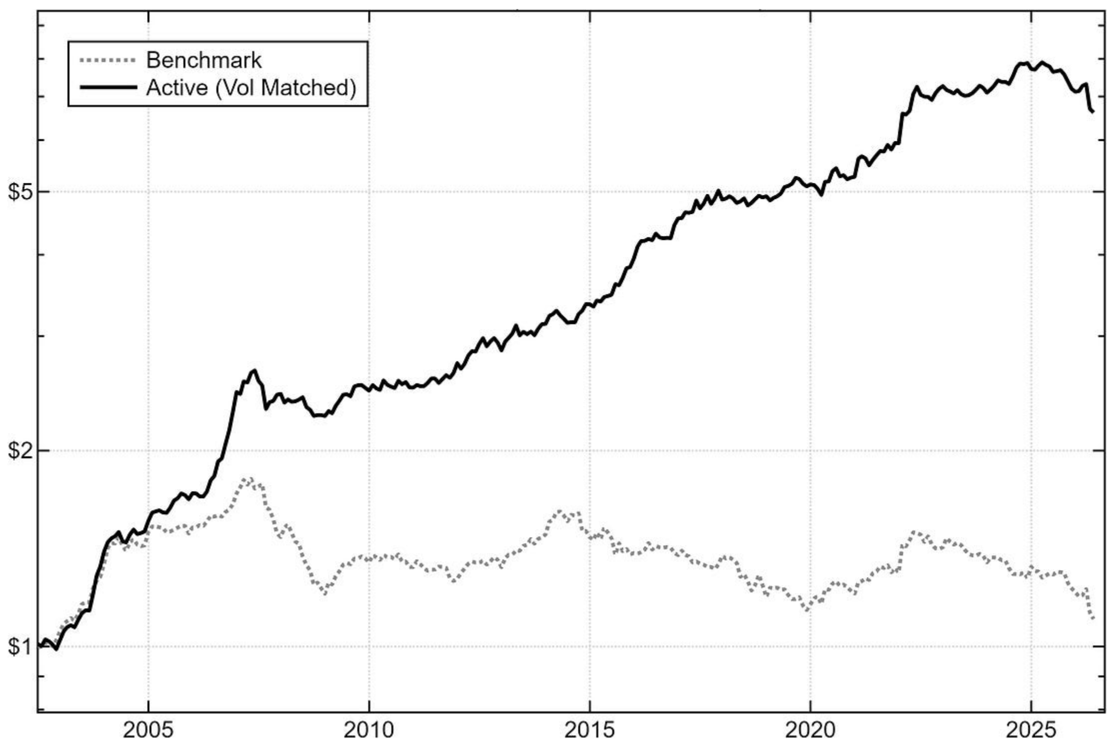
Implementing long-short equity portfolios presents several practical challenges, and although transaction costs remain an important consideration, today we specifically **focus on short-selling, which constitutes a significant (and less discussed) pain-point.**

In particular, we present an alternative implementation of our original study, built to offer a **more practical (and scalable) version** of the factors we previously examined.

Our results suggest that momentum effects persist under alternative factor-construction techniques that, over the last twenty years, would have still proven **effective in isolating exposure to the best-performing equity styles**.

Portfolio	CAGR	Volatility	Sharpe (0% Rf)	Max Drawdown
Benchmark	0.4%	7.6%	0.09	39.1%
Active	9.1%	8.4%	1.08	18.0%

As the numbers above highlight, **our actively managed framework helps revive an otherwise stagnant blend of factor portfolios.**



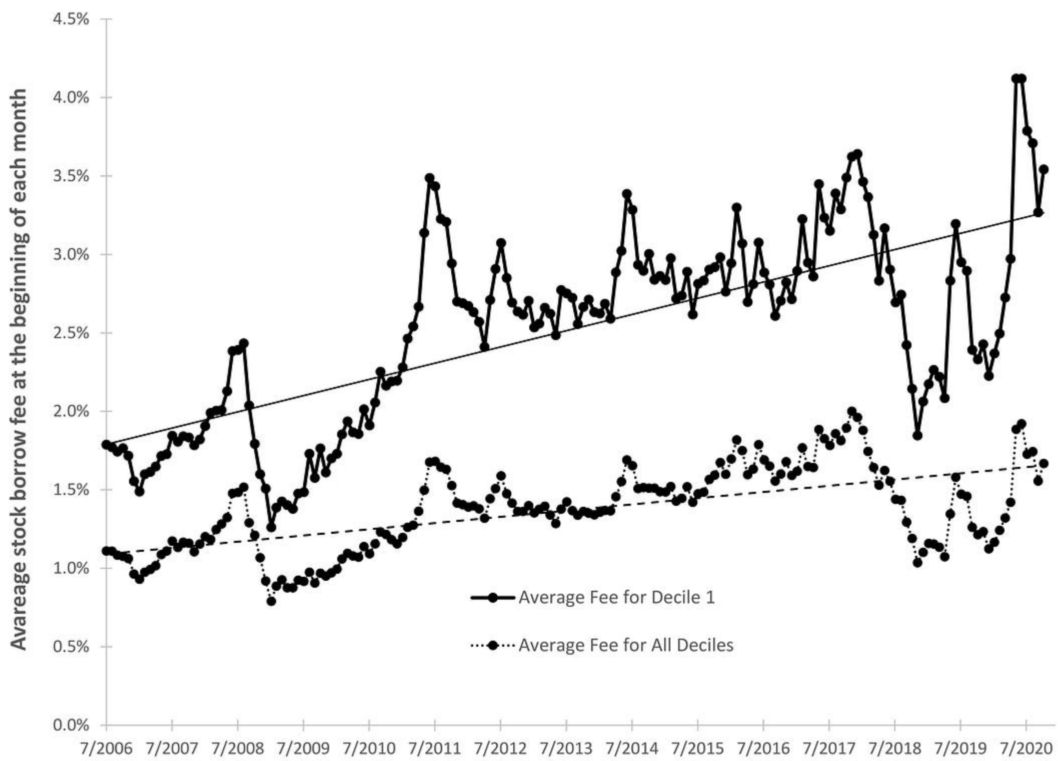
For investors seeking equity-style exposures that are **easier to trade in practice**, we believe this article offers some valuable insights.

An Alternative Factor Construction

As most traders will know, shorting individual names means accounting for **borrowing costs**, which represent the fee owed to whoever lends us the shares.

These costs are notoriously difficult to model and perhaps even harder to backtest, given that reliable securities-lending data only became available relatively recently and typically covers short samples.

For reference, Muravyev, Pearson, and Pollet (2024), in *“Anomalies and Their Short-Sale Costs”*, show how (across a comprehensive sample of 162 factor portfolios) the average long-short anomaly earns a *significant 0.14% per month before short-sale costs*, but essentially zero once stock borrow fees are deducted.



Average stock borrow fee at the beginning of each month (D1 and all deciles), [Anomalies and Their Short-Sale Costs (Muravyev, Pearson and Pollet, 2024)]

Crucially, **expensive-to-borrow** names are not randomly distributed across the market: they tend to cluster among smaller, less liquid stocks, which is precisely where many factors appear more effective (see Hou, Xue, and Zhang (2020) for additional discussion).

In our own work, we have also documented the tendency for thinner securities to amplify the effects of stylized behaviors and short-term signals. For more insights, see the links below.

Exploiting the Overnight Risk Premium

CONCRETUM RESEARCH · JUN 17

Read full story

Should You Trade Thin Stocks?

CONCRETUM RESEARCH · JUN 20

Read full story

Costs aside, there is also a matter of **share availability**: when we short shares we first need to find someone willing to lend them to us, and lendable supply is often constrained.

Needless to say, such issues become increasingly severe as AUM grows. Typically, institutional players try to mitigate them through **prime-brokerage relationships**: brokers can maintain extensive lending inventories across third-party lenders, and clients with more assets and higher trading volumes can negotiate meaningfully better locates. Yet even with these arrangements in place, the structural difficulty of running a large single-name short book remains, making pure long-short factor plays **quite hard to scale.**

One solution to work around these constraints could be to **operate at the factor-construction level**, attempting to harvest each factor premium without having to short individual names.

Following the logic of Blitz, Baltussen, and van Vliet (2020) in *"When Equity Factors Drop Their Shorts"*, this translates to maintaining a long position in the **D10** portfolio (the decile "bucket" expected to earn the documented premium) and hedging it with a **short broad-market exposure**: we will identify this framework as **D10-MKT**.

Equal- or Value-Weighted?

Before moving to the results, we need to address another important factor-construction choice.

For this study, we present the **equal-weighted** (EW) factor definition as our main specification rather than its **value-weighted** (VW) counterpart.

Generally speaking, VW portfolios are easier to scale, as they allocate more capital to larger, more liquid companies. Within a characteristic-sorted portfolio, however, value weighting also introduces a second (and **potentially unintended**) source of exposure. Once stocks are assigned to a decile, their weight is then determined by market capitalization: as a result, a handful of large-cap names can end up dominating portfolio risk, making decile-portfolio returns particularly sensitive to the idiosyncratic behavior of their largest constituents.

Equal-weighting, by contrast, assigns every stock in a decile the same weight and, for our purposes, offers a **cleaner expression of the characteristic used for ranking**.

Importantly, this does not make EW a free lunch: its heavier allocation to smaller stocks inevitably increases capacity constraints. That said, **in our implementation, these concerns are mitigated:** because D1 is replaced by a broad market hedge, we avoid having to short those thin securities that would otherwise make up the most demanding side of a canonical factor pair.

Building on our previous piece, we apply the construction outlined above while maintaining the same volatility-sizing approach. Both legs are sized to target **20% annualized volatility**, using a 12-month rolling standard deviation (σ) of (log) monthly returns as our volatility proxy:

$$w_{t-1} = \frac{20\%}{\sigma_{t-1}\sqrt{12}}$$

As a result, the monthly return (r_t) for each newly constructed (D10-MKT) factor formally becomes:

$$r_t^{\text{factor}} = w_{t-1}^{D10} r_t^{D10} - w_{t-1}^{MKT} r_t^{MKT}$$

Our universe of **fifteen anomalies** remains unchanged from the original study and it is summarized in the table below for continuity.

Factor	Description
Size	Market equity (price times shares outstanding). Long the smallest decile, short the largest.
Value	Book-to-market equity (BE/ME). Long the cheapest (high BE/ME) decile, short the most expensive.
Profitability	Operating profitability: revenues less costs and expenses, over book equity. Long the most profitable decile, short the least.
Investment	Growth of total assets over the prior fiscal year. Long conservative (low-investment) firms, short aggressive ones.
Momentum	Cumulative prior return, skipping the most recent month. Long recent winners, short recent losers.
Short-Term Reversal	Prior one-month return. Long recent losers, short recent winners.
Long-Term Reversal	Cumulative return from 60 to 13 months ago. Long long-term losers, short long-term winners.
Earnings / Price	Earnings-to-price ratio (E/P). Long the highest-E/P decile, short the lowest.
Cash Flow / Price	Cash-flow-to-price ratio (CF/P). Long the highest-CF/P decile, short the lowest.
Dividend Yield	Trailing dividend yield (D/P). Long the highest-yield decile, short the lowest.
Accruals	Change in operating working capital over book equity. Long the lowest-accrual decile, short the highest.
Beta	CAPM market beta of monthly returns. Long the lowest-beta decile, short the highest.
Net Share Issues	Growth in split-adjusted shares outstanding. Long the lowest-issuance decile, short the highest.
Variance	Variance of daily returns. Long the lowest-variance decile, short the highest.

Residual Variance	Residual variance of daily returns from the Fama-French three-factor model. Long the lowest decile, short the highest.

The fifteen decile-sorted anomalies used in the study. Descriptions adapted from the Kenneth French Data Library; returns are total returns (dividends included), not in excess of the risk-free rate.

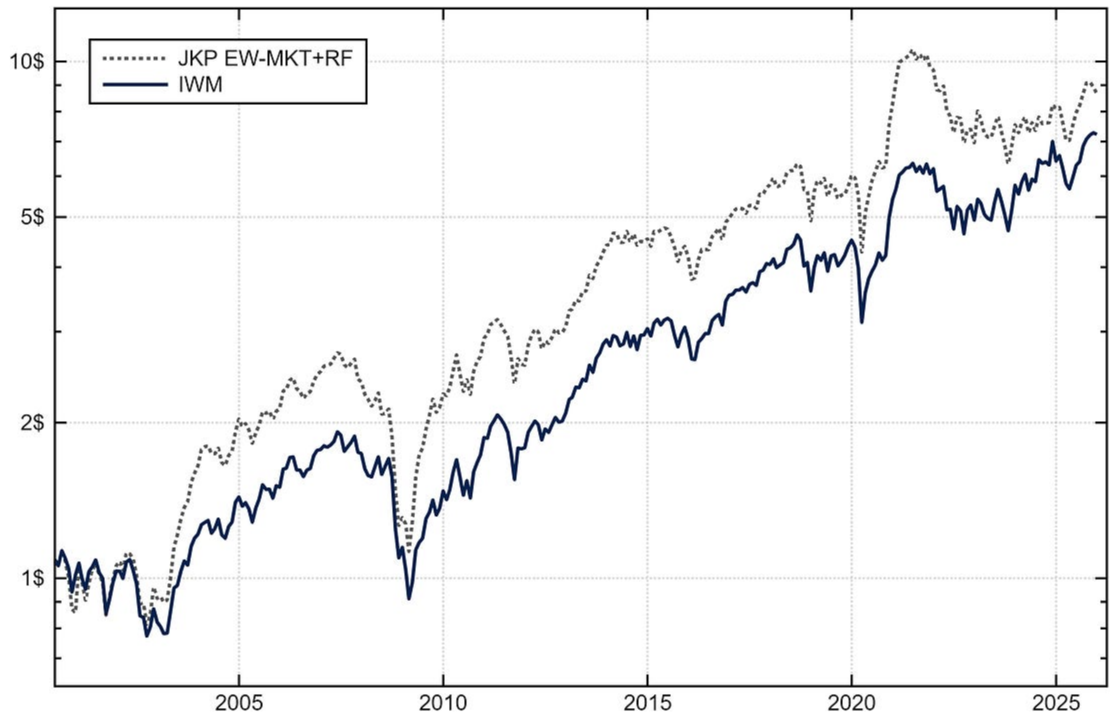
Choosing a Market Hedge

For our short-market leg (MKT), a suitable construction should retain the small-cap tilt naturally embedded in academic EW portfolios while remaining closely replicable via liquid index futures, as Blitz, Baltussen, and van Vliet (2020) themselves note.

As a reference point, we take the **US equal-weighted market portfolio (EW-MKT)** published in the Jensen, Kelly, and Pedersen data library (JKP), which covers the full US stock universe and applies equal weighting across companies, consistent with our EW factors. Unlike an ETF or futures contract, however, **it cannot be traded directly**.

In this context, the *Russell 2000 Inde*x represents a suitable candidate, given its exposure to the small-cap equity segment. In practice, it can be shorted through **IWM** (*iShares Russell 2000 ETF*) or, more efficiently, **through its futures contract** (RTY) **or micro-sized equivalent** (M2K), either of which offers a convenient, liquid, one-instrument hedge.

Comparing IWM's returns against those of the JKP EW-MKT portfolio, we find a correlation of 0.93 over their common history, which is why we select **IWM as our short-market proxy in every simulation that follows**. Accordingly, the D10-MKT specification will now be referred to as **D10-IWM**.



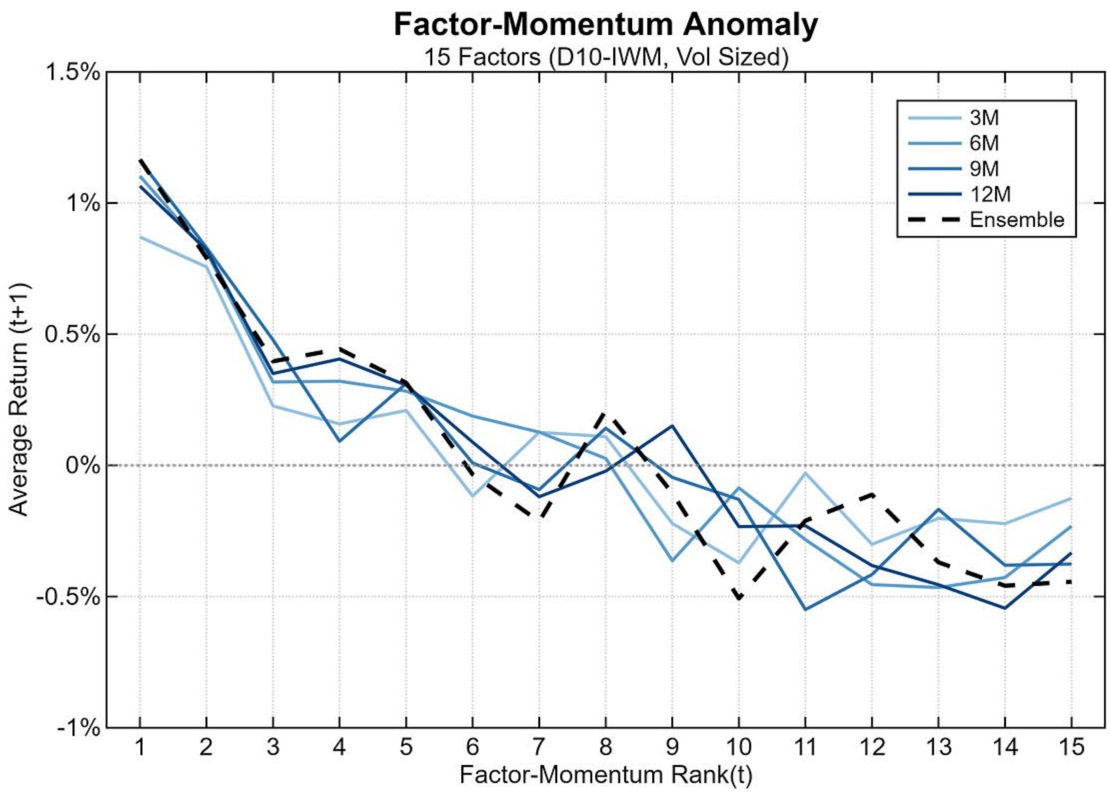
The JKP EW market is provided as an excess return series, therefore we add back the risk-free rate (RF) from the French dataset; correlations are computed from monthly returns over the common June 2000-December 2025 history.

Results

We begin by rerunning the same experiment from our first piece, as validation that this new factor methodology still exhibits **cross-sectional momentum** effects. At each month-end (*t*) we sort the fifteen D10-IWM factors by their trailing momentum (*M*), and for each rank we compute the average factor return in the following month (*t*+1).

$$M_{i,t}^{\mathrm{LB}} = \frac{P_{i,t}}{P_{i,t-\mathrm{LB}}} - 1, \qquad \mathrm{LB} \in \{3, 6, 9, 12\}$$

This time, we also expand the test across **four equally-spaced momentum lookbacks** (*LB* = 3, 6, 9 and 12 months), alongside an “ensemble” specification that ranks factors on the average of the four lookback-specific ranks. Our sample spans **June 2002** through **May 2026**, and the chart below reports the results:



Notably, all momentum lookbacks remain directionally consistent: average next-month returns overall decline as ranks move from 1 (the strongest relative-strength factor) to 15 (the weakest). As one might expect, **the picture is somewhat noisier than under the pure long-short factor construction** from our previous article, but the overall slope holds, which we read as additional confirmation of the cross-sectional momentum effect.

Next, we check whether any of the four momentum speeds consistently dominates the others. For every calendar year, we score each lookback by the average next-month return earned by its top three (monthly-selected) factors, and rank the four specifications from 1 (best) to 4 (worst).

Momentum	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	Avg
3M	3	3	3	1	2	2	4	1	4	4	1	4	2	4	4	1	4	4	1	3	4	3	1	4	1	2.7
6M	4	1	1	4	1	1	3	2	2	1	4	3	1	3	2	4	2	3	2	2	3	4	4	3	3	2.5
9M	1	2	4	3	3	4	1	3	1	3	2	1	3	2	3	2	1	2	3	1	2	1	2	1	4	2.2
12M	2	4	2	2	4	3	2	4	3	2	3	2	4	1	1	3	3	1	4	4	1	2	3	2	2	2.6

Yearly ranks of the 3, 6, 9 and 12-month momentum signals (1 = best, 4 = worst), scored by the average next-month return of each signal's top-three factors. The "Avg" column averages the ranks over the full sample.

As is rather common in quantitative modeling, **no single feature specification clearly and consistently trumps the others**: the 9-month signal leads with an average yearly rank of **2.2**, but the remaining specifications sit tightly between **2.5 and 2.7**. Moreover, this robustness to lookback selection is coherent with the findings of Gupta and Kelly (2019) in *"Factor Momentum Everywhere"*, who suggest that factors can be profitably timed using momentum formation windows anywhere from one month to five years.

With average ranks clustering closely across lookbacks, we adopt an ensemble approach, **sorting factors on their average momentum rank across the four specifications.**

A Combined Strategy

Having identified a directly tradable market proxy, we now turn to translating these above results into an implementable trading strategy. In particular we construct a **dual-momentum-type signal** (*DM*) **that layers a time-series component on top of our cross-sectional sort**.

> The rationale behind it is to gain exposure to the best-performing (D10-IWM) factors in *relative terms*, while still ensuring they are trending upward in *absolute terms*. We

consider this combination both economically sound and well grounded in the findings of our prior work.

The sorting operates exactly as outlined in the previous section: at each month-end, we rank all D10-IWM portfolios across four momentum speeds (3, 6, 9 and 12 months) and pick the three strongest (*TOP3*) based on their **average rank**.

Then, on top of this selection, we overlay a trend signal (*ts*): consistent with the cross-sectional features, it is built as the mean of four binary (0 or 1) trend indicators, one per lookback:

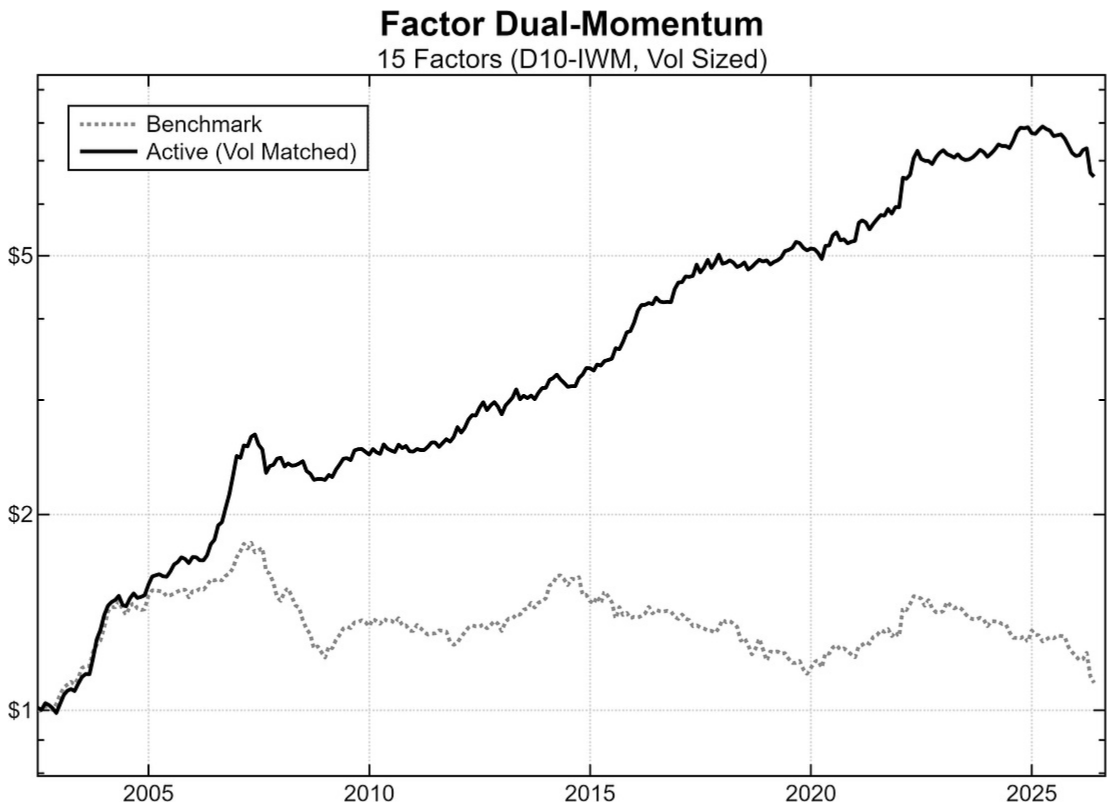
$$ts_{i,t} = \tfrac{1}{4}\sum_{\mathrm{LB}} \mathbf{1}\left\{M_{i,t}^{\mathrm{LB}} > 0\right\}, \qquad \mathrm{LB} \in \{3, 6, 9, 12\}$$

By construction, the combined indicator is **also bounded between 0 and 1**: it equals 1 when the factor is trending up on all four lookbacks, 0 when it is trending lower on all of them, and takes intermediate values otherwise.

Each of the three selected factors is allocated capital **in proportion to its trend signal** (*ts*), formally:

$$r_t^{DM} = \tfrac{1}{3}\sum_{i \in \mathrm{TOP3}_{t-1}} ts_{i,t-1} \cdot r_{i,t}$$

The chart below compares the dual-momentum strategy against a “passive” benchmark that holds all fifteen D10-IWM factors in equal weights and rebalances monthly.



For plotting, the dual-momentum strategy is volatility-matched to its benchmark.

Notably, the **passive basket delivers essentially flat performance**, whereas the active strategy compounds steadily and clearly outperforms over the available sample: from June 2002 to May 2026, the "always-long" portfolio posts a Sharpe ratio of just 0.09, compared with 1.08 for our active approach.

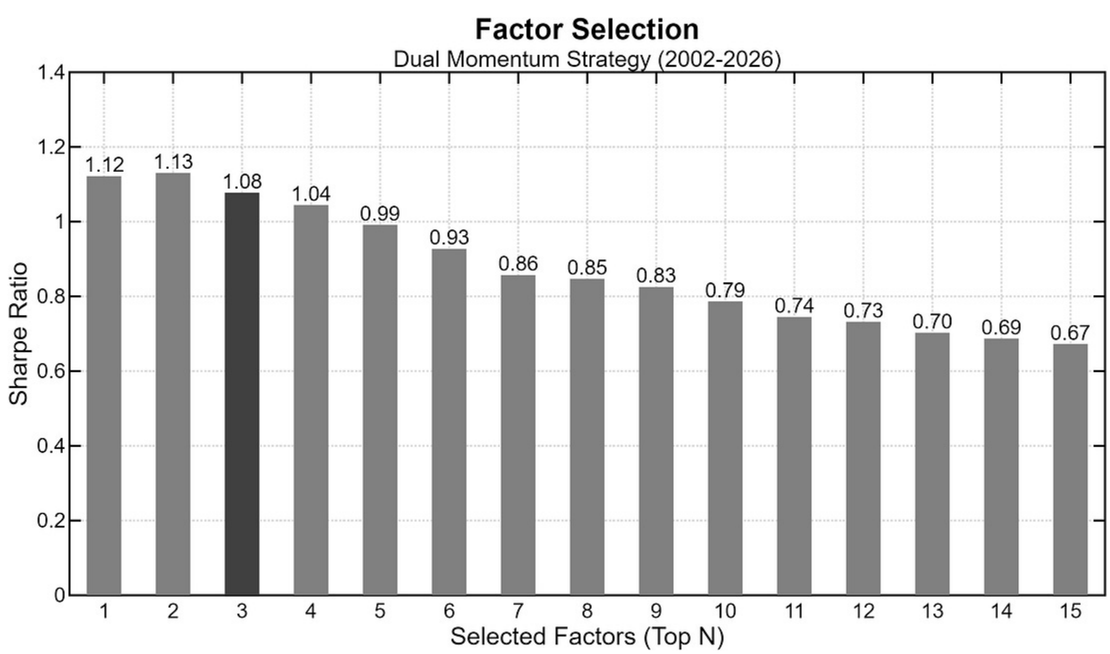
Portfolio	CAGR	Volatility	Sharpe (0% Rf)	Max Drawdown
Benchmark	0.4%	7.6%	0.09	39.1%
Active	9.1%	8.4%	1.08	18.0%

What we find most remarkable is that the dual-momentum framework effectively **revives an otherwise stagnant benchmark**: an investor passively holding all fifteen D10-IWM factors would have spent much of the sample in drawdown, whereas the momentum-and-trend overlay continued to perform at a respectable risk-adjusted pace.

These results also appear in line with our previous work, where the 2006-2026 period proved **favorable for both the cross-sectional and time-series momentum signals**, albeit under a more canonical and considerably harder-to-replicate factor construction.

Factor Selection

As a final robustness check, we examine how sensitive the full-sample results are to the number of factors selected by the *Top-N* momentum-rank, maintaining the time-series signal component throughout.



Sharpe ratios (0% Rf) as a function of selected top-N (D10-IWM) factors. The highlighted bar marks the top-three specification used throughout the article.

As the bar chart above highlights, risk-adjusted performance benefits from concentration in the strongest D10-IWM factors: Sharpe ratios decline almost monotonically as the book becomes more diluted, and holding all fifteen (with the trend-signal still active) nearly halves the Sharpe of the best specification. Notably, our choice (*N=3*) sits on the concentrated end of the spectrum, but **does not correspond to the in-sample optimum** (*N=2*).

Conclusion

All in all, our results confirm that **factor-momentum effects can survive alternative construction methodologies** deliberately designed to sidestep what is perhaps the biggest practical hurdle of equity investing: shorting individual names.

Notably, an ensemble of momentum speeds, combined with a trend overlay, proved effective even when the “passive” D10-IWM basket failed to earn meaningful returns.

Overall, the numbers suggest that *long-stock*, *short-broad-market* constructions represent a **promising area for further investigation**, not as an academic curiosity, but as a genuinely more implementable path to dynamically capturing equity-style premia.

This framework, which no longer involves shorting individual securities, **allows investors to push further into thinner names**, where factor effects tend to be largest. Still, we expect small-to-medium-sized funds or high-net-worth (HNW) individuals to be better positioned to exploit this (equal-weighted) approach than large institutions, whose scale makes meaningful allocations to thinner names considerably harder to execute. Trading costs would nonetheless warrant scrutiny, but they represent a more navigable issue than the operational limitations linked to shorting.

A final word of caution is in order: **academic sorts are excellent research instruments**, but before putting any factor-based strategy into production, we believe rebuilding the entire construction pipeline in-house can be a worthwhile step. This implies having control over the underlying data, both price and fundamental; liquidity and other "hygiene" filters; rebalancing policies, which can help mitigate *rebalance timing luck*; and the special treatment of certain asset classes (such as financials) among **many other delicate specification choices.**

For readers interested in taking this framework further (whether by testing alternative weighting schemes, extending it across geographies, or rebuilding it on different datasets) we welcome further discussion. Feel free to leave a comment, send us a direct message, or contact us at info@concretumgroup.com.

Research Transparency Statement

Every research idea, hypothesis, methodology, dataset, code, backtest, analysis, and conclusion presented in this article was independently developed by the Concretum Group research team. Artificial intelligence tools were used as productivity assistants to support the preparation of the final manuscript, including language refinement, grammatical review, and, where appropriate, the creation of tables, figures, and other visual elements. All research decisions, analyses, interpretations, and conclusions remain entirely those of the authors.